

MOUNT GAMBIER, AustraliaWMO: **948210**Lat: **37.7472S**Lon: **140.7739E**Elev: **69**StdP: **100.50**Time Zone: **9.50 (AUA)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.0	3.1	-1.2	3.4	19.0	0.5	3.9	12.0	11.9	10.2	10.8	11.4	1.6	30	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.7	34.7	18.0	31.7	17.5	28.9	17.0	20.0	28.5	19.0	27.2	18.1	26.0	6.4	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	12.9	21.7	16.6	11.9	20.4	15.6	11.2	19.6	57.7	28.3	54.3	27.2	51.3	26.2	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.9	9.1	-0.5	39.8	1.0	1.9	-1.2	41.2	-1.8	42.3	-2.3	43.4	-3.1	44.8		
(5)			-0.5	22.5	0.8	1.4	-1.1	23.5	-1.6	24.3	-2.1	25.1	-2.7	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	18.3	18.7	16.9	14.2	11.9	9.9	9.4	9.9	11.1	12.6	14.8	16.5		
	DBStd	4.44	3.95	3.88	3.61	2.93	2.36	1.91	1.76	2.04	2.44	3.23	3.68	3.72		
	HDD10.0	110	0	0	0	1	7	24	33	27	12	6	1	0		
	HDD18.3	1887	48	37	75	131	201	253	277	261	218	181	122	83		
	CDD10.0	1447	259	245	213	126	65	21	14	24	45	88	144	203		
	CDD18.3	181	48	49	29	6	1	0	0	0	1	5	16	27		
	CDH23.3	2391	659	571	351	76	3	0	0	0	3	72	237	418		
	CDH26.7	1101	338	276	154	19	0	0	0	0	0	21	93	200		
(14) Wind	WSAvg	4.5	4.9	4.8	4.4	3.9	4.0	4.1	4.6	4.7	4.8	4.7	4.7	4.9		
(15) Precipitation	PrecAvg	696	28	23	35	52	69	87	100	96	72	57	45	39		
	PrecMax	909	126	100	159	122	145	163	190	173	129	140	108	113		
	PrecMin	365	1	1	2	6	10	17	35	25	27	4	11	6		
	PrecStd	124	25	21	27	28	33	36	35	39	24	32	23	25		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.0	37.3	35.0	29.5	22.7	17.3	16.5	19.9	23.5	29.7	32.9	36.6		
		MCWB	18.5	18.9	17.6	15.6	13.5	12.9	11.7	12.4	14.4	15.8	17.4	17.1		
	2%	DB	34.7	33.7	31.0	25.4	19.4	15.4	14.7	17.2	20.0	25.1	29.5	32.2		
		MCWB	18.3	18.2	17.3	14.9	13.5	12.1	11.1	11.6	13.2	14.7	17.1	17.0		
	5%	DB	30.4	30.1	27.3	22.3	17.4	14.4	13.7	15.3	17.7	21.9	25.8	28.0		
		MCWB	17.6	17.9	16.3	14.4	13.1	11.6	10.6	11.1	12.3	14.0	16.3	16.7		
	10%	DB	26.5	26.4	23.8	19.7	15.9	13.4	12.7	13.8	15.9	19.0	22.3	24.2		
		MCWB	17.0	17.3	15.9	13.8	12.7	11.0	10.2	10.4	11.5	13.2	15.3	15.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.6	21.7	19.9	17.5	15.7	14.2	13.1	13.6	15.8	17.4	20.2	20.2		
		MCDB	29.7	29.2	28.3	23.4	18.1	15.6	15.0	17.7	20.2	24.5	27.3	28.1		
	2%	WB	20.0	19.7	18.5	16.3	14.7	12.9	11.9	12.4	14.1	15.7	18.4	18.7		
		MCDB	29.1	28.3	26.3	21.5	17.1	14.7	13.6	15.5	18.3	22.0	26.1	26.1		
	5%	WB	18.8	18.8	17.5	15.4	13.9	12.1	11.2	11.6	13.0	14.7	16.9	17.4		
		MCDB	27.2	27.1	24.2	20.0	16.2	13.7	12.8	14.3	16.4	20.6	23.8	25.7		
	10%	WB	17.6	17.7	16.5	14.5	13.1	11.4	10.6	10.9	12.1	13.5	15.5	16.2		
		MCDB	25.4	25.1	22.2	18.6	15.5	12.9	12.2	13.3	15.2	18.3	21.6	23.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.3	12.7	11.8	10.3	7.8	7.1	6.9	7.8	8.8	10.7	11.8	12.7		
		MCDBR	20.8	19.1	18.1	15.3	10.1	8.1	7.7	9.9	11.9	16.2	18.6	20.0		
		MCWBR	6.8	6.3	6.6	6.7	5.9	5.6	5.3	5.9	6.7	7.8	7.9	7.5		
	5% WB	MCDBR	16.5	15.6	14.4	11.8	8.1	7.0	6.9	8.7	10.1	14.1	16.1	16.5		
		MCWBR	6.8	6.2	6.3	6.2	5.4	5.3	5.1	5.6	6.3	7.5	7.8	7.2		
(40) Clear-Sky Solar Irradiance	taub		0.336	0.335	0.326	0.315	0.309	0.300	0.300	0.317	0.336	0.341	0.338	0.337		
	taud		2.524	2.539	2.561	2.578	2.570	2.586	2.582	2.521	2.457	2.447	2.483	2.506		
	Ebn at Noon		997	975	944	894	836	817	839	876	917	957	989	1001		
	Edh at Noon		110	104	94	81	71	64	69	84	102	113	114	113		
(44) All-Sky Solar Radiation	RadAvg		7.34	6.31	4.95	3.42	2.27	1.84	2.00	2.87	4.04	5.40	6.45	7.19		
	RadStd		0.39	0.32	0.23	0.22	0.12	0.12	0.11	0.14	0.32	0.41	0.38	0.46		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.37	N/A	N/A	N/A	N/A	+0.49	N/A	-115	+138	+40
(47)	Regional (22 neighbors)	+0.37	N/A	N/A	N/A	N/A	N/A	N/A	-96	+132	+39

Nomenclature: See separate page